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A group of students carried out an investigation to find the water potential of potato tissue.

The students were each given a potato and 50 cm³ of a 1.0 mol dm⁻³ solution of sucrose.

- They used the 1.0 mol dm⁻³ solution of sucrose to make a series of different concentrations.
- They cut and weighed discs of potato tissue and left them in the sucrose solutions for a set time.
- They then removed the discs of potato tissue and reweighed them.

Table 3 shows how one student presented his processed results.

Table 3

Concentration of sucrose solution / mol dm ⁻³	Percentage change in mass of potato tissue
0.15	+4.7
0.20	+4.1
0.25	+3.0
0.30	+1.9
0.35	– 0.9
0.40	– 3.8

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Explain why the data in **Table 3** are described as **processed** results.

[1 mark]

0 8 . **2** Describe how you would use a 1.0 mol dm^{-3} solution of sucrose to produce 30 cm^3 of a 0.15 mol dm^{-3} solution of sucrose.

[2 marks]

0 8 . **3** Explain the change in mass of potato tissue in the 0.40 mol dm^{-3} solution of sucrose.

[2 marks]

0 8 . **4** Describe how you would use the student's results in **Table 3** to find the water potential of the potato tissue.

[3 marks]

[Extra space] _____
